

STATISTICS PRACTICALS — QUICK REFERENCE NOTES

8 Topics | Proper Math Notation | What · When · Example · Formulas · Excel

1. Histogram, Average & Variance

What: Frequency bar chart + how spread out the data is

When: Continuous data — marks, heights, weights

Eg: Marks of 30 students, grouped in intervals of 10

FORMULAS:

$$\bar{x} = \frac{\sum x_i}{n}$$
$$\sigma^2 \text{ (population)} = \frac{\sum (x_i - \bar{x})^2}{n}$$
$$s^2 \text{ (sample)} = \frac{\sum (x_i - \bar{x})^2}{(n-1)}$$
$$\sigma = \sqrt{\sigma^2} \quad \text{(standard deviation)}$$

EXCEL:

=AVERAGE(range) [x̄]
=VAR.P(range) [σ² population]
=VAR.S(range) [s² sample]
=STDEV.S(range) [s]

2. Stem and Leaf Plot

What: Splits each value: tens digit = stem, units = leaf

When: Small dataset — see shape without losing values

Eg: 23 → stem=2, leaf=3 | 47 → stem=4, leaf=7

FORMULAS:

Stem = INT(value ÷ 10)
Leaf = value mod 10
Range = Max – Min
Median = middle value (sorted)
Mode = most repeated leaf

EXCEL:

=INT(A1/10) [stem]
=MOD(A1,10) [leaf]
=MEDIAN(range)
=MAX(range)-MIN(range)

3. Box Plot & Dot Plot

What: 5-number summary box — shows spread & outliers

When: Compare groups — which brand varies more?

Eg: Sales of Brand A vs B vs C side by side

FORMULAS:

Q1=25th % Q2=Median Q3=75th %
IQR = Q3 – Q1
Lower fence = Q1 – 1.5 × IQR
Upper fence = Q3 + 1.5 × IQR
Outlier: value < Lower OR > Upper

EXCEL:

=QUARTILE(range,1) [Q1]
=QUARTILE(range,3) [Q3]
=MEDIAN(range) [Q2]
IQR = Q3 – Q1 (manually)

4. Binomial Distribution

What: Counts successes in a FIXED number of trials

When: Only 2 outcomes: Success OR Failure (yes/no, H/T)

Eg: Flip coin 6 times. How many Heads? n=6, p=0.5

FORMULAS:

$$P(X=x) = C(n,x) \cdot p^x \cdot (1-p)^{n-x}$$
$$C(n,x) = \frac{n!}{x! \cdot (n-x)!}$$
$$\mu = n \cdot p \quad \text{(mean)}$$
$$\sigma^2 = n \cdot p \cdot (1-p) \quad \text{(variance)}$$

EXCEL:

=BINOM.DIST(x, n, p, FALSE)
→ P(exactly x successes)
=BINOM.DIST(x, n, p, TRUE)
→ P(x or fewer successes)

5. Poisson Distribution

What: Counts rare events in a fixed time or space

When: Random events at avg rate λ; no fixed n

Eg: Avg 2 accidents/day (λ=2). P(3 accidents)?

FORMULAS:

$$P(X=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$
$$\lambda \text{ (lambda)} = \text{average rate} = \text{mean}$$
$$e = 2.71828 \quad \text{(Euler's number)}$$
$$\mu = \lambda \quad \sigma^2 = \lambda \quad \text{(both equal } \lambda \text{)}$$

EXCEL:

=POISSON.DIST(x, λ, FALSE)
→ P(exactly x events)
=POISSON.DIST(x, λ, TRUE)
→ P(x or fewer events)

6. Regression Analysis (Y on X)

What: Fits best straight line: $\hat{Y} = a + bX$

When: Predict Y from X — does X cause change in Y?

Eg: X=study hours, Y=marks. b=5 → +1hr = +5 marks

FORMULAS:

$$b = \frac{n \sum XY - \sum X \cdot \sum Y}{n \sum X^2 - (\sum X)^2}$$
$$a = \frac{\sum Y - b \cdot \sum X}{n}$$

Line: $\hat{Y} = a + bX$
 $R^2 \rightarrow \text{closeness of fit (0 to 1)}$

EXCEL:

=SLOPE(Y_range, X_range) [b]
=INTERCEPT(Y_range, X_range) [a]
=FORECAST(new_x, Y_r, X_r) [Ŷ]
=RSQ(Y_range, X_range) [R²]

7. Correlation Coefficient — Pearson's r

What: Strength & direction of LINEAR link between X and Y

When: Do two variables move together? (study hrs & marks?)

Eg: r = 0.9 → strong positive: more study = more marks

Range: r always between -1 and +1

What does the r value mean?

r = +1 Perfect positive: X↑ then Y↑ always
0.7 to 0.99 Strong positive
0.4 to 0.69 Moderate positive
0.01 to 0.39 Weak positive
r = 0 No linear relationship
-0.39 to -0.01 Weak negative
-0.69 to -0.40 Moderate negative
-0.99 to -0.70 Strong negative
r = -1 Perfect negative: X↑ then Y↓ always

FORMULAS:

$$r = \frac{n \sum XY - \sum X \cdot \sum Y}{\sqrt{[(n \sum X^2 - (\sum X)^2) \cdot (n \sum Y^2 - (\sum Y)^2)]}}$$

Z-score method:

$$Z_x = (X - \bar{x}) / \sigma_x \quad Z_y = (Y - \bar{y}) / \sigma_y$$

$$r = \frac{\sum (Z_x \cdot Z_y)}{n - 1}$$

EXCEL:

=CORREL(X_range, Y_range) [r]
=PEARSON(X_range, Y_range) [r]
r > 0 → positive (both rise together)
r < 0 → negative (one rises, other falls)

8. Normal Distribution

What: Symmetric bell curve for continuous data

When: Heights, IQ, exam scores, measurement errors

Eg: IQ: μ=100, σ=15. Most people score 85-115

FORMULAS:

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$
$$Z = \frac{x - \mu}{\sigma}$$

μ ± 1σ → 68% of data
μ ± 2σ → 95% of data
μ ± 3σ → 99.7% of data

EXCEL:

=NORM.DIST(x, μ, σ, TRUE)
→ P(value ≤ x)
=NORM.INV(prob, μ, σ)
→ find x for given probability